Overview:
Ever since the Industrial Revolution, the predominant paradigm for manufacturing has been based on 'economies of scale' through repeatability, mass production and centralized processes. However, in the Internet age, new paradigms of production have emerged that 'democratize' and distribute the manufacturing of even complex products. Digital fabrication, online collaborative design and now AI-supported technologies are enablers of new production approaches that are based on 'economies of scope' where lots of different products, rather than the same product over-and-over, can be made profitably.

However, quality management has not yet adapted to this new digital, democratized, distributed reality. And now the biggest barrier to the adoption of distributed manufacturing is the challenge of distributed quality management. No open, shared, trusted, interoperable ecosystem currently exists to support democratized quality control and distributed quality assurance, verification, liability and continuous improvement across heterogeneous actors. Indeed, regulatory and certification frameworks have evolved to become significant and costly impediments even in traditional manufacturing, so innovation is needed.

Some disparate open-source products and approaches are emerging for distributed quality management. Examples were shared in the recent 'Distributed Quality Management' gatherings, such as the Global Distributed Tracking (GDT) tool from the Global Open-Source Quality Assurance System (GOSQAS). A community is forming from across open-source world, particularly from the Open Hardware movement.

This Phase 1 project will scope Open Distributed Quality Management (OpenDQM). It will scope an Open-Source Ecosystem (OSE) that operates community-governed infrastructures for stakeholders to collaboratively define, validate and steward quality practices specifically designed for distributed manufacturing environments. The project will map stakeholders and their needs, assess the market, report on the state of the art, engage regulatory and certification bodies to understand their challenges, prototype governance and data-exchange models and assess readiness for a Phase II implementation. It will use training on GDT to explore priorities and issues in this space – particularly with consumers and makers.

Intellectual Merits:
The project advances the science of distributed systems, quality management and socio-technical governance. It will develop the first 'state of the art' insight into diverse existing products and approaches and, by induction from these, develop a new typology for quality management. This will offer new architectures for federated quality assurance where many independent producers can share quality standards, testing protocols and verification data without needing costly centralized control. The project will contribute governance innovation for open technical ecosystems, likely more focused on stewardship than compliance. And the project will advance research into distributed manufacturing and resilient supply chains, particularly research into widespread PPE production in the COVID-19 response.

Broader Impacts:
The scope of the challenge in distributed quality management is enormous – just as it is in other areas that already have diverse paradigms of production, such as food (and, increasingly, news, music, movies and other industries affected by the Internet). This project will focus down that scope and its outputs will reach and impact: the Open Hardware movement; local and distributed manufacturers, and all the users and consumers who benefit from them, and; the quality management sector – particularly government regulators who struggle to stay ahead. It may also impact traditional manufacturers with new, open innovations in quality systems. Overall, massive numbers of small manufacturers could be positively impacted if the challenge of distributed quality management can be met, which would support new high-value jobs everywhere. And of course, it will ensure consumer, public and environmental safety in the new paradigm of production, where almost anyone can make almost anything almost anywhere.

Keywords: ENG; Quality Control; Quality Assurance; Safety; Manufacturing.